

Deep Learning-Based Detection and Segmentation of Fetal Heart Components in Prenatal Echocardiograms

Malek Ben Yaala

Medical Imaging Technology Lab-LTIM, Monastir University, Tunisia

benyaala.maleke@gmail.com
(Corresponding author)

Rim Zarrouk

Medical Imaging Technology Lab-LTIM, Monastir University, Tunisia
rima.zarrouk@gmail.com

Ramzi Mahmoudi

Medical Imaging Technology Lab-LTIM, Monastir University, Tunisia

ramzi.mahmoudi@esiee.fr

Abstract—Monitoring the fetal heart during gestation is crucial for early detection of cardiac abnormalities. However, manual screening can present several challenges due to the poor quality of echocardiographic images, which exhibit high levels of darkness, as well as variations in fetal position and inter-operator variability.

In order to help clinicians diagnose fetuses, this study suggests an automated approach based on deep learning techniques. In order to identify, categorise, and segment various fetal cardiac structures, such as the left atrium (LA), left ventricle (LV), right atrium (RA), right ventricle (RV), and aorta in prenatal echocardiograms, we employ the Mask R-CNN architecture with ResNet101 as its foundation. With a mean average precision (mAP) of 86.60% for object detection and 88.25% for mask generation, a mean intersection over union (mIoU) of 67.88%, a mean Dice coefficient of 76.54%, a mean recall of 75.80%, a mean precision of 79.18%, and a mean area under the ROC curve (AUC) of 87.71%, the results demonstrate the model's strong ability to recognise and segment these anatomical structures. Our model outperformed several existing methods, which indicates that it has the ability to serve as a computer-aided tool for clinicians during prenatal screening.

Index Terms—Deep learning, Mask R-CNN, segmentation, object detection, medical imaging, ultrasound, echocardiography, fetal heart.

I. INTRODUCTION

Congenital abnormalities affecting the fetal heart remain among the leading causes of neonatal complications and mortality worldwide. These conditions, known as congenital heart diseases (CHDs), are considered a primary cause of perinatal mortality. The mortality rate among live births affected by CHDs is around 12%, and 25% of newborns with CHD die few weeks after birth. [1]. This makes prenatal screening essential for detecting abnormalities at an early stage and deciding the appropriate interventions at the right time to reduce the perinatal mortality rate and save fetal lives. Prenatal screening is performed using medical imaging techniques, particularly ultrasound imaging, which allows visualization of the fetal heart and its different components, including the left and right cavities, vessels, septa, and the aorta [2].

The main problem faced in prenatal screening is that echocardiograms present shadows and artifacts caused by the necessity to penetrate two different bodies to reach the fetal heart. This complicates the interpretation of the images, which can lead to errors in manual diagnosis.

Recent advances in artificial intelligence, particularly deep learning (DL) approaches, have significantly improved automatic medical image analysis to address the mentioned challenges and improve diagnosis accuracy [3]. Particularly, convolutional neural networks (CNNs) provide automatic processing of different types of images, including medical images and echocardiograms. They help precisely detect and delineate the boundaries of different substructures within medical images, assisting clinicians in identifying any potential structural abnormalities, such as atrophy or hypertrophy of the heart cavities, or even pathologies affecting vessels or aorta. In the present study, we propose an approach based on CNNs for automatic echocardiogram processing. Our study aims to introduce a computer-aided diagnosis (CAD) tool for prenatal screening. In this work the main contributions can be summed up as follows:

- We propose a multi-task approach for simultaneous detection, classification, and segmentation of cardiac substructures of the fetal heart in prenatal ultrasound images;
- We target five cardiac anatomical structures, including the left atrium (LA), left ventricle (LV), right atrium (RA), right ventricle (RV), and the aorta;
- We collect data from various sources to enable the model learn patterns from images acquired using multiple acquisition machines, thereby improving model generalization and enhancing applicability in different real screening scenarios;
- We used medical imaging metrics like mean average precision (mAP), mean intersection over union (mIoU), mean Dice similarity coefficient (DSC), recall, precision, and area under the ROC curve to evaluate the model's

performance on unseen data.

The rest of the paper is organized as follows: Section II discusses earlier efforts, while Section III describes the dataset and preprocessing processes. Section IV describes the model and methodologies, while Section V summarizes the results of computer simulations and analysis.

II. RELATED WORKS

DL techniques have exhibited considerable utility in automating tasks such as the detection and segmentation of anatomical structures in medical ultrasound imaging. These techniques addressed traditional challenges such as poor image quality and variability between different operators. Nagabotu et al. [4] used the U-Net architecture to segment the fetal head in ultrasound images. Meanwhile, Singh et al. [5] (2025) employed the U-Net++ architecture with a ResNet-34 backbone combined with attention blocks to segment the fetal brain. Several research works specifically addressed fetal cardiac analysis in prenatal echocardiograms, including the study introduced by Rachmatullah et al. [6] (2021) for semantic segmentation of the fetal heart as a whole and the study proposed by Stoean et al. [7] (2022) for the delineation of eleven individual cardiac structures within the fetal heart, both using the U-Net architecture. Other studies have utilized two cascaded U-Nets (CU-Net) to segment seven cardiac substructures within the fetal heart [8] (2020). Another study focused on the detection and segmentation of the ventricular septum by combining YOLOv2 with U-Net and VGG-16 [9] (2020). Other variants of the YOLO architecture, including YOLOv7, were used for the real-time detection of nine cardiac substructures [10] (2023). Additionally, Yan et al. [11] (2024) proposed a framework that combines the detection of regions of interest with a multi-scale feature extraction module based on the attention mechanism for segmentation of the three-vessel plane in fetal echocardiography and differentiation of small vessel structures. Hurtado et al. [12] (2024) introduced a hybrid approach that integrates multiple models to segment different anatomical structures in 2D fetal heart ultrasound images of various qualities. Li et al. [13] (2025) developed a Dense Atrous Spatial Pyramid Pooling (Dense ASPP) model with convolutional block attention modules (CBAM) to segment the apical four-chamber view and address challenges such as speckle noise and anatomical variability.

III. DATASET AND PREPROCESSING

This study utilized ultrasound data acquired from the apical four-chamber (4CH) view, a cardiac imaging plane that depicts the heart chambers: the LA, LV, RA, RV, and aorta. The dataset was collected from three different sources, including Chinese hospitals [14], a publicly available FOCUS dataset [15], and the Radiopaedia website [16]. The Chinese dataset consists of 24 2D ultrasound images in PNG format, obtained from the First and Second Hospitals of Fujian Medical University, China. The echocardiograms were acquired during the second and third trimesters of gestation. The FOCUS dataset consists of 299 fetal ultrasound images showing the cardiac and

thoracic regions of fetuses in the second trimester of gestation. FOCUS was publicly published as part of open science to contribute to various research areas, mainly fetal diagnosis and the detection of congenital heart disease. The data collected from the Radiopaedia website consist of ultrasound videos and images of fetuses in the second trimester of gestation. The multi-source dataset helps the model learn patterns from images acquired from multiple machines, which prevents dependence on a single acquisition type.



Fig. 1. Sample of raw data collected from Chinese hospitals, FOCUS dataset, and the Radiopaedia website (from left to right)

The data collected were preprocessed using multiple techniques to prepare them for the training phase. Ultrasound videos were extracted into different frames using the OpenCV library. Together with the ultrasound images, this resulted in a total of 332 ultrasound images. The frames obtained were then cleaned by removing redundant and corrupted images. For a single model input, each image was downsized to 512x512 pixels. The images were further augmented with the Contrast Limited Adaptive Histogram Equalisation (CLAHE) technique to augment local contrast and boost the visibility of substructures inside the fetal heart, hence aiding the model in precisely delineating the borders of each object. The images were then annotated using the CVAT tool under the guidance of medical experts. Then binary masks were generated to serve as ground truth for both training and evaluation. To enlarge the dataset, several data augmentation techniques were employed, including rotation, shearing, contrast adjustment, cropping, and zoom out. Following data augmentation, the total number of images increased to 2,734. The obtained data were divided into three main sets in proportions of 80%, 10%, and 10%. The training set contained 2,188 images, while both validation and test sets comprised 273 images. To avoid overfitting and reliance on augmented data or on one acquisition type, we ensured that each set contains both augmented and original images as well as samples from each dataset. Fig 2 presents the different steps of the preprocessing phase.

IV. METHODOLOGY

In this study, we utilised the Mask R-CNN architecture for the concurrent detection, classification, and segmentation of five anatomical components within the fetal heart: the LA, left ventricle, right atrium, right ventricle, aorta. Mask R-CNN consists of four core modules: a backbone, a region proposal network (RPN), a RoI Align module, and a multi-task module with three parallel heads [17]. The backbone is a CNN that is responsible for automatic feature extraction from input images.

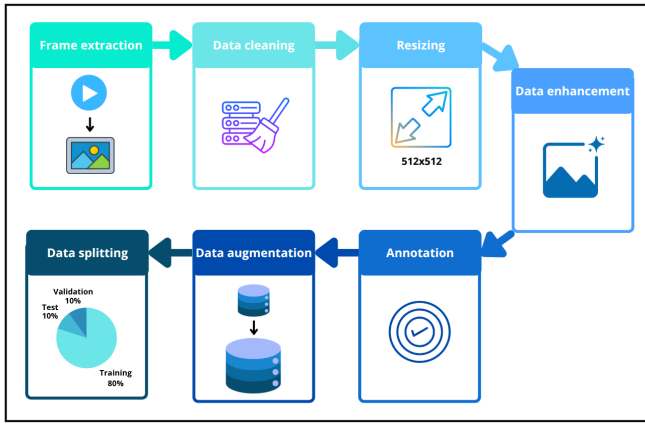


Fig. 2. Preprocessing steps for the multi-source dataset

In our study, we used ResNet101 [18] and feature pyramid network (FPN) [19] as the backbone. This backbone was pretrained on the Common Object in Context (COCO) dataset, where it learned to delineate boundaries of objects with various shapes and sizes. Then it was fine-tuned using our dataset of fetal echocardiograms to adapt its features to the specific domain. The second module of Mask R-CNN is the RPN, which takes as input the extracted feature maps and generates candidate regions of interest (RoIs), which are areas with a high probability to contain target objects. The RoI align module then aligns the RoIs on the extracted feature maps, using bilinear interpolation, to obtain RoI features. The final module contains three parallel heads: classification, bounding box, and mask heads. The classification head predicts class probabilities of objects in the generated RoIs using fully connected (FC) layers with a SoftMax layer for final class prediction. The bounding box head refines the coordinates of RoIs for precise localization. Finally, the mask head segments the detected objects and generates binary masks using a fully convolutional network (FCN), enabling pixel-wise segmentation.

Mask R-CNN is a two-stage object detection architecture. The first stage consists of proposing candidate RoIs that might contain target objects. The second is the multi-task processing of those regions, including classification, bounding box regression, and segmentation. The Mask R-CNN architecture and the different stages are illustrated in Fig 3.

The input to the Mask R-CNN is an image with the representation (W, H, C), where W is the width, H is the height, and C is the number of channels which is generally 3 for RGB representation. In our case, when dealing with ultrasound frames, the input images are converted to grayscale to reduce the complexity of images, which makes the model focus on the intensity distribution rather than colors. This makes it easier for the model to detect the heart cavities and the aorta by focusing on dark regions, where the intensity is significantly lower compared to other pixels. The grayscale image is fed to the backbone for feature extraction, using ResNet101 and FPN. The ResNet101 uses multiple convolutional blocks with

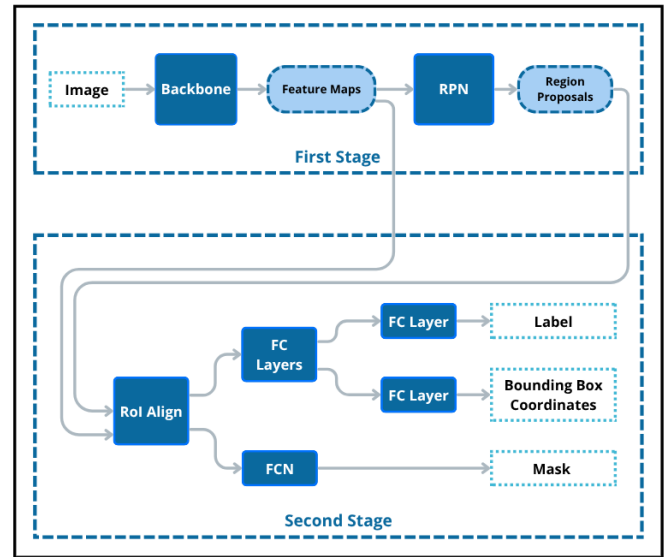


Fig. 3. The two-stage Mask R-CNN architecture

several kernels. Each kernel allows for the detection of a specific characteristic, such as edges, corners, shapes, or textures. The output of this step is a multi-channel representation of the image. Each channel indicates the activation map of the feature detected by the specific filter. The activation map shows where and how strongly a certain feature appears in the image.

The produced feature maps are then passed through the FPN to produce feature pyramids that help with handling objects of varying sizes. The pyramid includes feature maps with low spatial resolution and stronger semantics at higher levels, and high spatial resolution but weaker semantics at lower levels. The resulting feature maps have the same number of channels (256) but differ in spatial resolution (1/4, 1/8, 1/16 and 1/32 of the input image size), corresponding to different levels of the pyramid. This generates multi-scale feature maps from a single image, which enables the detection of objects with varying sizes.

RPN is a neural network that uses convolutional layers to slide over the feature maps obtained from the backbone and generate multiple anchors, which are predefined bounding boxes with different scales and aspect ratios. Then these anchors are passed through two convolutional layers. The first layer, with the SoftMax activation function, predicts the objectness score of each anchor and specifies whether the contained object belongs to the background or to the foreground. The second layer performs bounding box regression to refine anchor coordinates towards actual object boundaries. The output of the RPN is a set of regions of interest (RoIs), which are candidate locations that have a high probability of containing target objects.

The RoI Align (Region of Interest Align) takes two inputs: the first is the feature map, which is a rich representation of the image produced by the backbone, and the second is the RoIs generated by the RPN, which indicate the coordinates

of the areas on which the network should focus. These RoIs have coordinates that represent boxes with varying sizes. The RoI Align obtains the coordinates of these RoIs and aligns them on the feature maps to extract the representation inside those boxes, and then it resizes them to produce fixed-size (generally 7x7) feature maps. Unlike RoI Pool, which performs rounding and quantization of the obtained coordinates while resizing, which causes misalignment, the RoI Align uses a technique called bilinear interpolation that preserves exact spatial correspondence. This accurate alignment is important to ensure that features passed to the detection and the mask heads are spatially well-aligned, improving localization and segmentation accuracy.

The classification and bounding box heads together perform object detection, which was the main role of Faster R-CNN [20]. Mask R-CNN extends this by adding a new branch for mask generation. This branch separately performs semantic segmentation using a fully convolutional network (FCN). Semantic segmentation assigns a class label to each pixel, producing a dense classification map where all pixels that belong to the same semantic category have the same label (color). In mask R-CNN, this segmentation is applied to each RoI obtained after object detection. By combining these two concepts and performing semantic segmentation for each detected object, Mask R-CNN achieved instance segmentation, where each individual object is precisely segmented at the pixel level.

Mask R-CNN performs parallel tasks that are optimized end-to-end during training by a multi-task loss function. This enables the model to accurately execute various tasks. The overall loss function comprises the summation of the classification loss, the bounding box regression loss, and the mask loss. The classification loss is the cross-entropy loss that quantifies the mistake in the given category for each identified object. The bounding box regression loss is the smooth L1 loss that measures the discrepancy between the predicted coordinates of the detected object and the matching ground truth coordinates. The mask loss is the pixel-wise binary cross-entropy loss that quantifies the disparity between the anticipated mask and the ground truth mask.

We configured the training to 50 epochs, at which point the model attained stability. We employed a learning rate of 0.0001 and a momentum of 0.9 utilizing the stochastic gradient descent (SGD) optimizer. The model underwent end-to-end optimization with a multi-task loss function and was trained on a P100 GPU with 16 GB of RAM.

V. RESULTS AND DISCUSSION

To evaluate the performance of our model, we used multiple metrics such as mean average precision (mAP), dice similarity coefficient (DSC), intersection over union (IoU), recall, precision, and Area under ROC Curve (AUC). These metrics are commonly used to quantify the model's output in the context of medical imaging. During validation, the model was optimized to improve the precision of both detection and segmentation while ensuring having a decreasing loss function

value to avoid overfitting. The model reached a stable state after 50 epochs, where precision was no longer improved for multiple iterations. Table I presents the model's evaluation in the stable state. It achieved an mAP of 88.62% for bounding box regression and 91.36% for mask generation. The mean IoU was 71.97% and the mean DSC was 79.64%. The model reported a mean recall of 79.76%, a mean precision of 81.53%, and an overall AUC of 89.66%.

TABLE I
RESULTS OF THE COMPREHENSIVE EVALUATION OF OUR MODEL
DURING VALIDATION

Structure	mAP/BBox (%)	mAP/Segm (%)	IoU (%)	DSC (%)	Recall (%)	Precision (%)	AUC (%)
RV	87.34	90.67	76.02	83.82	80.50	88.20	90.12
LV	93.12	96.98	72.92	81.09	78.64	85.08	89.11
RA	93.10	93.10	72.37	78.59	77.57	80.47	88.66
LA	88.94	90.65	73.33	81.06	82.58	80.90	91.11
Aorta	80.64	85.41	65.21	74.76	78.94	73.03	89.34
Mean	88.62	91.36	71.97	79.76	79.64	81.53	89.66

When evaluated on unseen data, as presented in Table II, the model demonstrated a strong ability to detect five different objects within the fetal heart, achieving a mean AP (mAP) of 86.60% for the bounding box regression. The evaluation of the model's capability to segment the detected object resulted in an mAP of 88.25%, a mean DSC of 76.54%, and a mean IoU of 67.88%, proving the strong overlap between the predicted masks and the corresponding ground truth mask. This highlights the model's ability to accurately delineate the boundaries of the cardiac structures in ultrasound images. The mean recall and precision achieved 75.80% and 79.18%, respectively, indicating that the model is able to effectively detect relevant instances and avoid misclassifications. The mean AUC across all classes reached 87.71%, demonstrating the model's ability to correctly distinguish between the different classes belonging to the foreground.

TABLE II
RESULTS OF THE COMPREHENSIVE EVALUATION OF OUR MODEL
DURING TEST

Structure	mAP/BBox (%)	mAP/Segm (%)	IoU (%)	DSC (%)	Recall (%)	Precision (%)	AUC (%)
RV	80.50	82.18	66.88	77.19	73.06	84.82	86.35
LV	92.01	91.75	63.98	71.90	72.81	72.40	85.94
RA	94.01	94.01	71.71	80.10	80.31	81.51	89.89
LA	88.17	90.11	58.92	69.32	67.36	73.72	83.86
Aorta	78.31	83.23	77.92	84.21	85.47	83.45	92.55
Mean	86.60	88.25	67.88	76.54	75.80	79.18	87.71

Among the analyzed structures, the aorta produced comparatively lower detection scores than the cardiac chambers, with 78.31%, for bounding box regression. This may be explained by the fact that the aorta is the smallest detected object compared to the heart chambers. Despite the data enhancement techniques employed, small vascular substructures like the aorta may still be affected by speckle and shadowing artifacts in ultrasound imaging, making them difficult for the model to detect. The LV and RA achieved the highest metrics given that they have typical shapes and clear cavity boundaries, which improve their visibility and facilitate detection. In contrast, the RV has a coarse texture due to the existence of extra structures

like trabeculations and the moderator band (MB) inside it, that can obscure cavity boundaries in some frames. Similarly, the LA often has an irregular shape due to the vascular structures adjacent, which can make it difficult for the model, in some cases, to identify the exact limits of the LA.

Fig. 4 illustrates the qualitative evaluation of the model and presents prediction samples of purely non-augmented test images from both the Radiopaedia dataset, in row (a), and the FOCUS dataset, in row (b). All predicted masks are comparable to the corresponding ground truth masks. The confidence scores of all identified structures are over 90%, which further proves the strong capacity of our model to accurately detect and segment the heart cavities as well as the aorta in fetal echocardiograms. The experimental findings additionally indicate that the augmentation strategy contributed positively to model generalization. The model successfully learns patterns from both augmented and non-augmented images during training and demonstrates high prediction performance for purely non-augmented samples of different acquisition types during inference, proving its strong generalization ability.

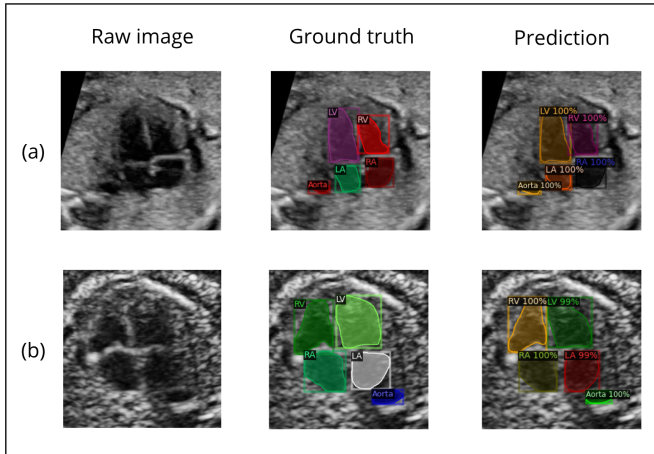


Fig. 4. Samples of non-augmented test images from (a) the Radiopaedia dataset and the (b) FOCUS dataset, with predicted masks corresponding to anatomical structures of the fetal heart in prenatal echocardiogram: LA, LV, RA, RV, and aorta.

For benchmarking, we compared our model with existing methods specialized in the detection of fetal heart components and with others focusing on segmentation tasks. The first category, presented in Table III, includes SONO, a method proposed by Komatsu et al. [21] and designed to detect eighteen cardiac substructures of the fetal heart, including the left and right cavities, as well as veins, vessels, and other anatomical structures. SONO achieved an overall AP (mAP) of 70.20%.

Similarly, the study introduced by Iriani Sapitri et al. [10] targeted the detection of nine anatomical structures of the fetal heart, achieving an mAP and an overall AUC of 82.10%. When compared to these methods, our model based on Mask R-CNN and ResNet101 demonstrated higher performance in detecting different components of the fetal heart, including the chambers

of the fetal heart and the aorta, with an mAP of 86.60% and an overall AUC of 87.71%.

TABLE III
RESEARCH BENCHMARKS WITH OTHER APPROACHES FOR THE
DETECTION OF CARDIAC SUBSTRUCTURES

Author	Method	Medical object	mAP (%)	AUC (%)
Komatsu et al. [52]	SONO (Darknet19-based YOLOv2)	18 cardiac substructures	70.20	-
Iriani Sapitri et al. [10]	Darknet53-based YOLOv7	9 cardiac substructures	82.10	82.10
Our proposed model	ResNet101-based Mask R-CNN	5 cardiac substructures	86.60	87.71

Existing methods that focused on the segmentation task are summarized in Table IV. This includes the study proposed by Dozen et al. [9] that targeted the ventricular septum of the fetal heart and achieved an IoU of 68.91% and a DSC of 55.43%. Yang et al. [22] segmented the fetal heart using U-Net++, resulting in a limited IoU value of 48%. Nurmaini et al. [23] targeted the segmentation of the fetal heart in prenatal and postnatal ultrasound images, achieving a moderate IoU of 55.72%, a DSC of 42.39%, and an mAP of 51.04%. In contrast to the mentioned studies that segmented the fetal heart as a whole, our model was able to delineate the boundaries of five individual structures within the fetal heart, generating a per-object binary mask for each category. Our model achieved an overall IoU of 67.88%, a mean DSC of 76.54%, and a high mAP value of 88.25%. The results highlight the improved performance of our model compared to the existing methods.

TABLE IV
RESEARCH BENCHMARKS WITH OTHER APPROACHES FOR SEGMENTATION
OF CARDIAC SUBSTRUCTURES

Author	Method	Medical object	IoU (%)	DSC (%)	mAP (%)
Dozen et al. [9]	YOLOv2, U-Net and VGG-16	Ventricular septum	68.91	55.43	-
Yang et al. [22]	U-Net++	Heart	48	-	-
Nurmaini et al. [23]	ResNet101	Heart	55.72	42.39	51.04
Our proposed model	ResNet101-based Mask R-CNN	5 cardiac substructures	67.88	76.54	88.25

Although our model reports higher evaluation metrics than other existing methods, the comparison results may raise some reliability concerns, since approaches employ different evaluation setups and datasets and also perform distinct tasks. The present study still has some limitations: (i) the model was trained on a relatively small dataset, acquired from a single imaging view (apical 4CH), (ii) cross validation could be applied to ensure the robustness of the model, and (iii) incorporating other fetal cardiac planes than the apical 4CH could improve generalizability and broaden the scope of applicability.

VI. CONCLUSION

Fetal cardiac screening is a critical problem that necessitates more attention and efforts to reduce prenatal mortality. Deep learning-based approaches can facilitate the screening process by assisting pediatricians and radiologists in fetal diagnosing. In this context, we proposed a Mask R-CNN-based model

that demonstrates a strong capability in identifying the heart chambers and the aorta of the fetal heart in ultrasound images. The model was able to delineate the boundaries of these anatomical structures, providing a visualization of the shapes of different components of the fetal heart in the 2D apical four-chamber view. This helps clinicians analyze the structural appearance of the fetal heart components and detect possible pathologies. We aim to conduct clinical validation to ensure the reliability of our model in real-world scenarios and to demonstrate its utility in assisting clinicians during prenatal screening.

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